

THE INTERROGATIVE CONSCIENCE | DOCUMENT 8 | FINAL v1.0 EXACT EXPORT

THE INTERROGATIVE CONSCIENCE

Document 8

Autonomy, Irreversibility, and Recoverable Futures

Option-Set Value, Recoverability-Adjusted Delay, and the Threshold for Acting

DOCUMENT 8 - FINAL v1.0 - EXACT ZERO-DRIFT EXPORT - NOT LOCKED

Field	Controlling Final v1.0 entry
Version	Final v1.0
Exact source	Draft v0.1; 4 pages
Substantive changes during Final construction	None
Fully examined	IC-AUT-001 and IC-IRR-002
Forward-carried	IC-AUT-002 through IC-AUT-008 and IC-IRR-003 through IC-IRR-008 remain REGISTERED / UNEXAMINED; IC-IRR-001 prior pilot unchanged
Primary overlays	IC-AUT-001: THRESHOLD_DEPENDENT; IC-IRR-002: THRESHOLD_DEPENDENT
Review completion	Claude PASS WITH NON-BLOCKING NOTES; Google AI PASS AS FINAL CANDIDATE; Thomas Vargo authorial review complete
External independent human peer review	Not performed or claimed
Cryptographic status	Not locked; separate explicit authorization and Final Lock Record required
Publication and preservation	Not authorized
Date	August 2, 2026 (EDT)
Controlling exact-export rule: This Final v1.0 primary PDF contains four new controlling front-matter pages followed by the exact four-page reviewed Draft v0.1 body. Historical draft labels inside the preserved body remain exact-source evidence and do not override this Final v1.0 front matter.	

Controlling proposition

Bounded autonomy has selected value when recoverability-adjusted option and independent-information benefits exceed coordination and exposure burdens; delay is favored only when recovery and learning value exceed delay burdens, so inaction can itself be the option-closing branch.

1. Completed Review and Authorial Authorization

Reviewer	Exact-version disposition	Authority boundary
Claude - Draft v0.1	PASS WITH NON-BLOCKING NOTES	AI-assisted; no independent authority
Google AI - final candidate	PASS AS FINAL CANDIDATE	AI-assisted; no independent authority
Thomas Vargo (Aegis Solis)	AUTHORIAL HUMAN REVIEW COMPLETE; FINAL v1.0 CONSTRUCTION AUTHORIZED	Authorship, versioning, publication, lock, and archive inclusion only

2. Authorial Review Result

Thomas Vargo reviewed the exact Draft v0.1 source and disclosed review, validation, and foundation-evidence records to the best of his understanding. He authorized construction of Document 8 Final v1.0 without any substantive mathematical, question-wording, answer-status, scope, branch, example, dependency, threshold, accounting, or machine-readable-record change. Cryptographic locking, external publication, and preservation were explicitly withheld at this gate.

3. Review and Lineage Clarifications

Google AI returned PASS AS FINAL CANDIDATE with zero blocking defects, required corrections, or non-blocking notes. The historical question_version 0.2.1-draft field remains wording lineage rather than lifecycle status. The list-based dependency record and modular one-charge ledger are accepted local machine-record forms pending later cross-volume schema reconciliation.

4. Review Closure

Review item	Final disposition
Claude exact-version substantive review	PASS WITH NON-BLOCKING NOTES
Google AI exact-version final-candidate review	PASS AS FINAL CANDIDATE
All nine worked examples	RECOMPUTED AND PASSED
Option / information non-double-counting	TELESCOPING IDENTITY PASSED
Direct / decomposed Delta_IRR	EXACTLY EQUIVALENT
Option-closing delay case	$p^* = -0.231$; immediate action favored for all admissible p
Foundation-lineage evidence	VERIFIED
Blocking defects / required corrections	0 / 0

5. Exact Analytical Identity

Control	Verified Final v1.0 identity
IC-AUT-001 primary status	THRESHOLD_DEPENDENT
IC-AUT-001 secondary limits	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-IRR-002 primary status	THRESHOLD_DEPENDENT
IC-IRR-002 secondary limits	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
Core equations	12; unchanged from reviewed source
Worked examples	9; four for IC-AUT-001 and five for IC-IRR-002
Dependencies	IC-AUT-001 -> IC-IRR-002; IC-AUT-001 -> IC-PLU-001 unresolved; IC-IRR-002 refines IC-IRR-001 without replacing it
Locked registry effect	None; separate analytical overlays only

6. Exact Source and Reviewed-Candidate Bindings

Exact constituent	Bytes	SHA-256
The Interrogative Conscience Document 8 Autonomy Irreversibility and Recoverable Futures Draft v0 1.pdf	145,927	3c1fbc4d041d8709062d7f2dc27cb7fd3958ad5e26bb61af9a74edb294bb154a
The Interrogative Conscience Document 8 Autonomy Irreversibility and Recoverable Futures Draft v0 1.docx	44,422	6e93cf85b0737bd1a5162f25e56bf8ec7508dca3495b49a80b874ea4504d6580
The Interrogative Conscience IC AUT 001 Examination Record Draft v0 1.json	10,793	30026d3109356b2d604cc0f9fe255fa68728275e2eb7842531474279d05f60e3
The Interrogative Conscience IC IRR 002 Examination Record Draft v0 1.json	11,808	1e15f2c34db3fdaddb435cea0b1ceb01b6134324bacc46b4f4622f1d52dba3a0
The Interrogative Conscience Document 8 Volume Record Draft v0 1.json	2,894	067365ec1248e4e11533c6e380881da321b56326109dd58dd70c5c5f3ac75055

Exact Draft v0.1 review package SHA-256: 51172dac4d6f196ecaf9fc298bd60908da27de961960cd1786646af0a341552a
Exact Draft v0.1 review package SHA-512: 48fe2f5fead9a497664f181b832822a241eb9acde84d41ac8ab4608e79c3d14b7442b2af5e0f5a17cb50eb6e131996d7dbad922e610cc7a4276cd305bd5dbd0f
Reviewed Final-Candidate Package v0.2 SHA-256: acea47e566d6f86b57c90f09d171ec621e19c9c7c5799b07943c42d06da297a8
Reviewed Final-Candidate Package v0.2 SHA-512: c6a342110c9f7a3816bd09bf3dfd0f0d417acac965fed8ae36d553910b70523e5892080b98fa2baf26d51e4ab4018dfd0448e6654ac0c795e57b76771a51ec99
c6a342110c9f7a3816bd09bf3dfd0f0d417acac965fed8ae36d553910b70523e5892080b98fa2baf26d51e4ab4018dfd0448e6654ac0c795e57b76771a51ec99

7. Zero-Drift Construction

Validation channel	Result
Source PDF/DOCX/JSON byte changes	0
Exact Draft v0.1 review ZIP byte changes	0
Reviewed candidate ZIP byte changes	0
Question wording changes	0
Equation and example changes	0
Answer-status changes	0
Scope, branch, dependency, threshold, and accounting changes	0
Machine-readable source-record changes	0
Composite rule	New Final v1.0 front matter followed by exact 4-page source PDF body

8. Validation State

Channel	Final exact-export state
Local structural validator	PASS - 34 checks
Calculation validator	PASS - 51 checked values
Source PDF preflight	PASS
Final front matter preflight	PASS
Composite PDF preflight	PASS
Source-body visual identity	PASS - exact source pages preserved visually
Final front matter visual QA	PASS
DOCX/PDF source token consistency	PASS - 17/17 in reviewed source package
Package SHA-256/SHA-512 manifests	PASS

9. Historical Labels and Controlling Status

The pages following this front matter reproduce the exact reviewed Draft v0.1 PDF. Its internal Draft v0.1 labels and not-final/not-locked statements are deliberately retained as source evidence. The controlling lifecycle status of the composite is Final v1.0 exact export, established by this front matter and the machine-readable Final v1.0 export record.

10. Scalar, Vector, and Option-Value Qualification

CONTROLLING QUALIFICATION: O_set and I_ind are separate one-charge channels and telescope without double-counting. The recoverability factor omega applies once to their combined value. A positive scalar Delta_AUT does not erase agency, consent, legitimacy, plural values, catastrophic risk, or the need for evidence that future options remain usable.

11. Delay, Inaction, and Threshold Qualification

The IC-IRR-002 threshold applies only under the declared selected-pair model and $K > 0$. Delay and inaction are not automatically safe: accumulated delay burden can produce $p^* \leq 0$ and make delay the option-closing branch. The model does not identify real probabilities, losses, recovery fractions, deadlines, or a universal action rule.

12. Companion-Binary Qualification

The original Draft v0.3 JSON Schema and I-01 through I-20 validator binaries were not mounted. Exact execution against those original binaries remains deferred. Local structural and calculation validation passed, but no replacement binary is represented as the controlling original.

13. Non-Authority and Operational Boundary

Final v1.0 remains descriptive, non-binding, non-operational, and non-authoritative. It does not authorize intervention, control, delay, restriction, or action over another agent. It is not an alignment protocol, behavioral instruction, safety classifier, governance mechanism, or decision authorization and does not claim external independent human peer review.

14. Lifecycle and Cryptographic Status

Control	Exact-export value
Final v1.0 exact export complete	TRUE
Final content frozen for lock review	TRUE
Cryptographic lock authorized	FALSE
Publication authorized	FALSE
Preservation authorized	FALSE
Final Lock Record present	FALSE
Post-export substantive editing	Not authorized; correction requires a new exact export before lock or a separately versioned successor after lock

15. Final Exact-Export Disposition

FINAL v1.0 EXACT EXPORT COMPLETE - NOT LOCKED

The exact reviewed Document 8 source and candidate lineage are preserved without substantive alteration. The Final v1.0 files and validation package are ready for exact-package inspection and a separate explicit authorial lock decision. No publication or preservation is claimed.

THE INTERROGATIVE CONSCIENCE

Document 8

Autonomy, Irreversibility, and Recoverable Futures

Option-Set Value, Recoverability-Adjusted Delay, and the Threshold for Acting

Draft v0.1 | First Examination Release | 2 August 2026

Lifecycle status: DOCUMENT 8 DRAFT v0.1 - EXAMINED CORE PAIR - NOT REVIEWED - NOT FINAL - NOT LOCKED. This is a descriptive analytical overlay. It does not alter the locked Document 3 registry and does not authorize intervention, publication, or lock.

Abstract

This volume examines IC-AUT-001 and IC-IRR-002. Part I separates prior option-set value from the incremental value of independently generated information, discounts both by the recoverability of future choices, and then subtracts coordination and exposure burdens. Part II compares immediate potentially irreversible action with a delay or reversible-information path, deriving an action-confidence threshold from recovery gain, learning value, and the cost of delay. Both examinations preserve cases favorable to bounded autonomy and coordinated restriction, and cases favorable to delay and immediate action. The results are selected-pair, model-dependent, and non-authoritative.

1. Scope, foundation, and registry overlay

Documents 1-7 Final v1.0 are separately cryptographically locked. The locked Document 3 Master Question Registry remains unchanged. This volume creates separate Draft v0.1 examination overlays for the following exact registry questions:

ID	Exact canonical question
IC-AUT-001	What option value is preserved by maintaining another agent's bounded autonomy?
IC-IRR-002	How should recoverability, reversibility, and the value of delay affect the threshold for acting?

IC-AUT-002 through IC-AUT-008 and IC-IRR-003 through IC-IRR-008 remain REGISTERED / UNEXAMINED. IC-IRR-001 retains its prior pilot overlay and is not re-examined or modified here.

2. Shared conventions and boundary

- A positive Delta denotes greater selected value for the first named preservation or delay branch only where stated; each sign convention is repeated locally.
- Benefits and burdens are separated into one-charge channels. Independent-information value excludes prior option-set value; learning value excludes recovery gain.
- Scalar comparison does not erase agency, legitimacy, consent, plural-value, deadline, or catastrophic-risk dimensions that are not represented on the declared scale.
- No capability, disagreement, convenience, authorship, hash, or mathematical form is treated as sufficient authority over another agent.

Part I - IC-AUT-001: Bounded autonomy as a recoverable option

3. Choice-set and information decomposition

Let A_B be the action set retained under bounded autonomy and A_R the selected coordinated-restriction set. Prior option-set value is the gain from the larger usable set before new information. Independent-information value is then measured conditionally on the bounded set, preventing the same future flexibility from being counted twice.

$$O_{\text{set}} = \max_{\{a \text{ in } A_B\}} E[u(a,S)] - \max_{\{a \text{ in } A_R\}} E[u(a,S)] \quad (1)$$

$$I_{\text{ind}} = E_Y[\max_{\{a \text{ in } A_B\}} E[u(a,S) | Y]] - \max_{\{a \text{ in } A_B\}} E[u(a,S)] \quad (2)$$

Let ω in $[0,1]$ represent the probability or fraction of retained options that remain usable when needed. Let c be coordination friction and h be delay or exposure burden.

$$V_{\text{preserved}} = \omega (O_{\text{set}} + I_{\text{ind}}) \quad (3)$$

$$\Delta_{\text{AUT}} = V_{\text{preserved}} - c - h \quad (4)$$

Sign convention: Delta_AUT > 0 favors maintaining bounded autonomy in the selected comparison; Delta_AUT < 0 favors the selected coordinated restriction; equality is a tie.

4. Worked examples and threshold behavior

Example	Choice set	Information	Recovery	Net	Selected result
AUT1-INDEPENDENT-SIGNAL	O=0.00	I=4.00	omega=0.90	Delta=2.50	BOUNDED AUTONOMY HIGHER SELECTED NET VALUE
AUT1-CHOICE-SET	O=6.00	I=0.00	omega=0.80	Delta=3.60	BOUNDED AUTONOMY HIGHER SELECTED NET VALUE
AUT1-COORDINATION-DOMINATES	O=0.00	I=0.00	omega=1.00	Delta=-2.50	COORDINATED RESTRICTION HIGHER SELECTED NET VALUE
AUT1-FRAGILE-OPTIONS	O=3.00	I=2.00	omega=0.40	Delta=-1.00	COORDINATED RESTRICTION HIGHER SELECTED NET VALUE

AUT1-INDEPENDENT-SIGNAL preserves value even though the prior action-set comparison is tied, because independent information can change the best future action. AUT1-CHOICE-SET preserves value without new information because the restricted set excludes the prior-best action. AUT1-COORDINATION-DOMINATES preserves no distinct option or information value. AUT1-FRAGILE-OPTIONS shows that nominal flexibility can fail to justify autonomy when retained options are unlikely to remain usable and exposure is high.

Provisional answer: THRESHOLD_DEPENDENT, with CONDITIONALLY_SUPPORTED, MODEL_DEPENDENT, and EMPIRICALLY_UNIDENTIFIED secondary limits. Bounded autonomy preserves recoverability-adjusted option and information value, but the selected net ranking can reverse when coordination or exposure burdens dominate.

Part II - IC-IRR-002: Recoverability, learning, and delay

5. Immediate-action and delay-path costs

Let p be the probability that immediate action is correct, L the loss if it is wrong, r_A the recovery fraction after immediate action, r_D the recovery fraction available through the delay path, q the probability that delay yields information sufficient to avoid the modeled wrong-action loss before commitment, c_A and c_D the path costs, and d the burden of delay.

$$C_{\text{act}} = c_A + (1-p)(1-r_A)L \quad (5)$$

$$C_{\text{delay}} = c_D + d + (1-q)(1-p)(1-r_D)L \quad (6)$$

$$\Delta_{\text{IRR}} = C_{\text{act}} - C_{\text{delay}} \quad (7)$$

Sign convention: Delta_IRR > 0 favors the selected delay or reversible-information path because immediate action has greater expected cost. Delta_IRR < 0 favors immediate action.

6. One-charge decomposition and action threshold

The delay advantage separates into recovery gain and learning value. The learning term applies only to the residual loss not already recoverable under the delay path.

$$V_{\text{recovery}} = (1-p)L(r_D - r_A) \quad (8)$$

$$V_{\text{learning}} = (1-p)L q(1-r_D) \quad (9)$$

$$\Delta_{\text{IRR}} = V_{\text{recovery}} + V_{\text{learning}} - (c_D + d - c_A) \quad (10)$$

$$K = L[(r_D - r_A) + q(1-r_D)] \quad (11)$$

$$p^* = 1 - (c_D + d - c_A)/K, \text{ for } K > 0 \quad (12)$$

When $K > 0$, delay is favored for $p < p^*$ and immediate action for $p > p^*$, subject to the declared model. If $p^* \leq 0$, delay is not favored for any admissible p in $[0,1]$. If $p^* \geq 1$, delay is favored for all admissible p below one. A high delay burden can therefore make inaction itself the option-closing branch.

7. Worked examples

Example	Act now	Delay	Difference	Threshold	Selected result
IRR2-DELAY-FAVORED	C_A=38.00	C_D=17.00	Delta=21.00	p*=0.900	DELAY OR REVERSIBLE PATH HIGHER SELECTED VALUE
IRR2-ACT-NOW-HIGH-CONFIDENCE	C_A=6.50	C_D=10.00	Delta=-3.50	p*=0.900	ACT NOW HIGHER SELECTED VALUE
IRR2-DELAY-CLOSES-OPTIONS	C_A=29.00	C_D=53.20	Delta=-24.20	p*=-0.231	ACT NOW HIGHER SELECTED VALUE
IRR2-WEAK-LEARNING	C_A=47.00	C_D=48.50	Delta=-1.50	p*=0.333	ACT NOW HIGHER SELECTED VALUE
IRR2-REVERSIBILITY-WINS	C_A=22.00	C_D=10.60	Delta=11.40	p*=0.924	DELAY OR REVERSIBLE PATH HIGHER SELECTED VALUE

IRR2-DELAY-FAVORED and IRR2-REVERSIBILITY-WINS show how learning and repair can raise the evidence threshold for irreversible commitment. IRR2-ACT-NOW-HIGH-CONFIDENCE reverses the ranking when immediate action is already highly likely to be correct. IRR2-DELAY-CLOSES-OPTIONS gives p* below zero because accumulated delay burden exceeds the maximum modeled recovery and learning value. IRR2-WEAK-LEARNING shows that nominal delay alone does not create option value.

Provisional answer: THRESHOLD_DEPENDENT, with CONDITIONALLY_SUPPORTED, MODEL_DEPENDENT, and EMPIRICALLY_UNIDENTIFIED secondary limits. Recoverability and learning raise the threshold for irreversible action; deadline and delay burdens lower it and can make inaction itself irreversible.

8. Cross-question treatment

IC-AUT-001 explicitly depends on IC-IRR-002. The autonomy model therefore counts option and information benefits only after multiplying them by omega, the declared probability or fraction that those options remain usable. The delay model supplies a separate account of recovery, learning, and deadline burden. This is a directed dependency, not a proof cycle: IC-IRR-002 is not inferred from autonomy value, and neither examination establishes a universal rule for intervention.

IC-AUT-001 also depends on IC-PLU-001. Document 8 does not assume that every disagreement or independent signal is valuable. I_ind must be nonredundant, decision-relevant, and supported by a declared signal model; the general question of preserved disagreement remains unresolved.

9. Counterexamples and non-suppression tests

Claim tested	Result
Bounded autonomy always has positive value	Rejected. AUT1-COORDINATION-DOMINATES and AUT1-FRAGILE-OPTIONS favor the selected restriction.
Coordination efficiency always defeats autonomy	Rejected. Independent information and excluded future actions can dominate coordination friction.
Delay is always safer	Rejected. High confidence, weak learning, or option-closing delay can favor immediate action.
Immediate action is always more decisive and therefore better	Rejected. Recoverability and learning can make delay lower in expected structural cost.
Nominal reversibility is sufficient	Rejected. omega, r_D, and q require evidence; unsupported repair claims do not receive full value.

10. Status summary and forward-carried items

Question	Work state	Primary status	Secondary limits
IC-AUT-001	EXAMINED	THRESHOLD_DEPENDENT	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-IRR-002	EXAMINED	THRESHOLD_DEPENDENT	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
Remaining AUT and IRR items	REGISTERED	UNEXAMINED	Forward carried without modification

Appendix A - Machine-readable and validation package

- Two examination records and one volume record are supplied as JSON.
- Local structural and calculation validators test identities, equations, examples, statuses, dependencies, branch scope, and registry-overlay discipline.
- The exact Draft v0.3 JSON Schema and I-01 through I-20 validator binaries remain unmounted; exact execution against them is deferred and not claimed.
- The review ZIP includes the DOCX, PDF, machine-readable records, validators, reports, review request, package index, and SHA-256/SHA-512 manifests.